

Causal inference is not just a statistics problem

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**Causal Inference is not a
statistics problem**

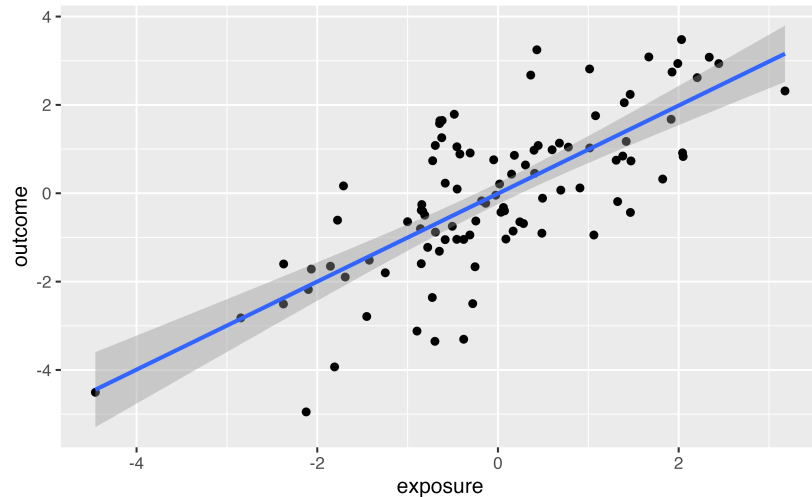
Causal Inference is not
just a statistics problem

The problem

We have measured variables, what should we adjust for?

exposure	outcome	covariate
0.49	1.71	2.24
0.07	0.68	0.92
0.40	-1.60	-0.10
.	.	.
.	.	.
.	.	.
0.55	-1.73	-2.34

What does the data say?



```
1 cor(exposure, covariate)
```

```
[1] 0.7
```

The exposure and measured factor are positively correlated

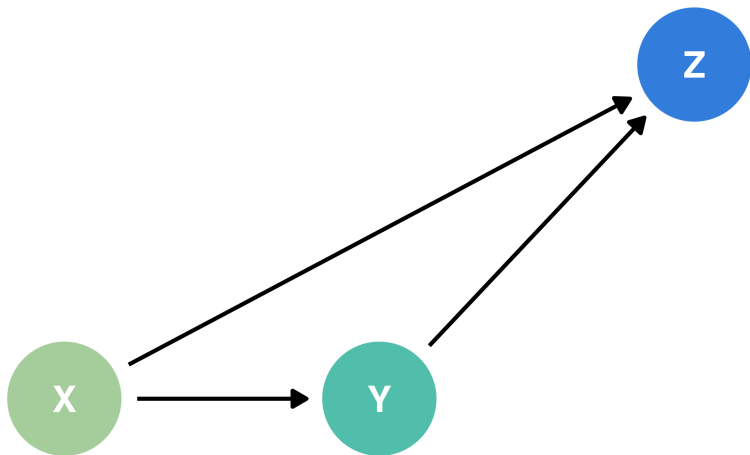
One unit increase in the exposure yields an average increase in the outcome of 1



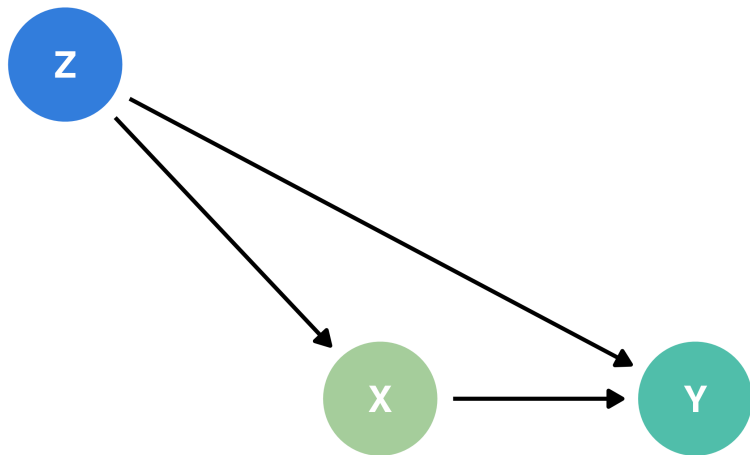
**To adjust or not adjust?
That is the question.**

Causal Quartet

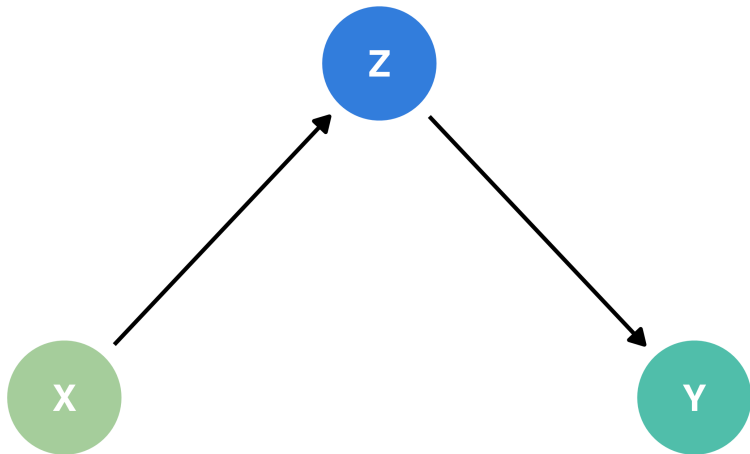
(1) Collider



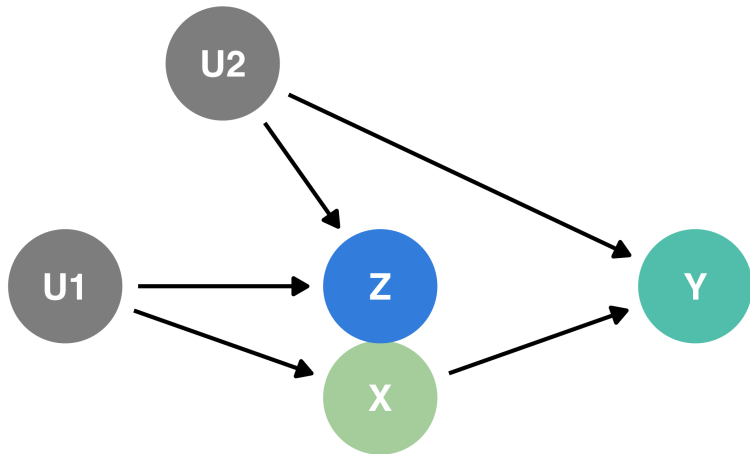
(2) Confounder



(3) Mediator



(4) M-bias





Your turn 1

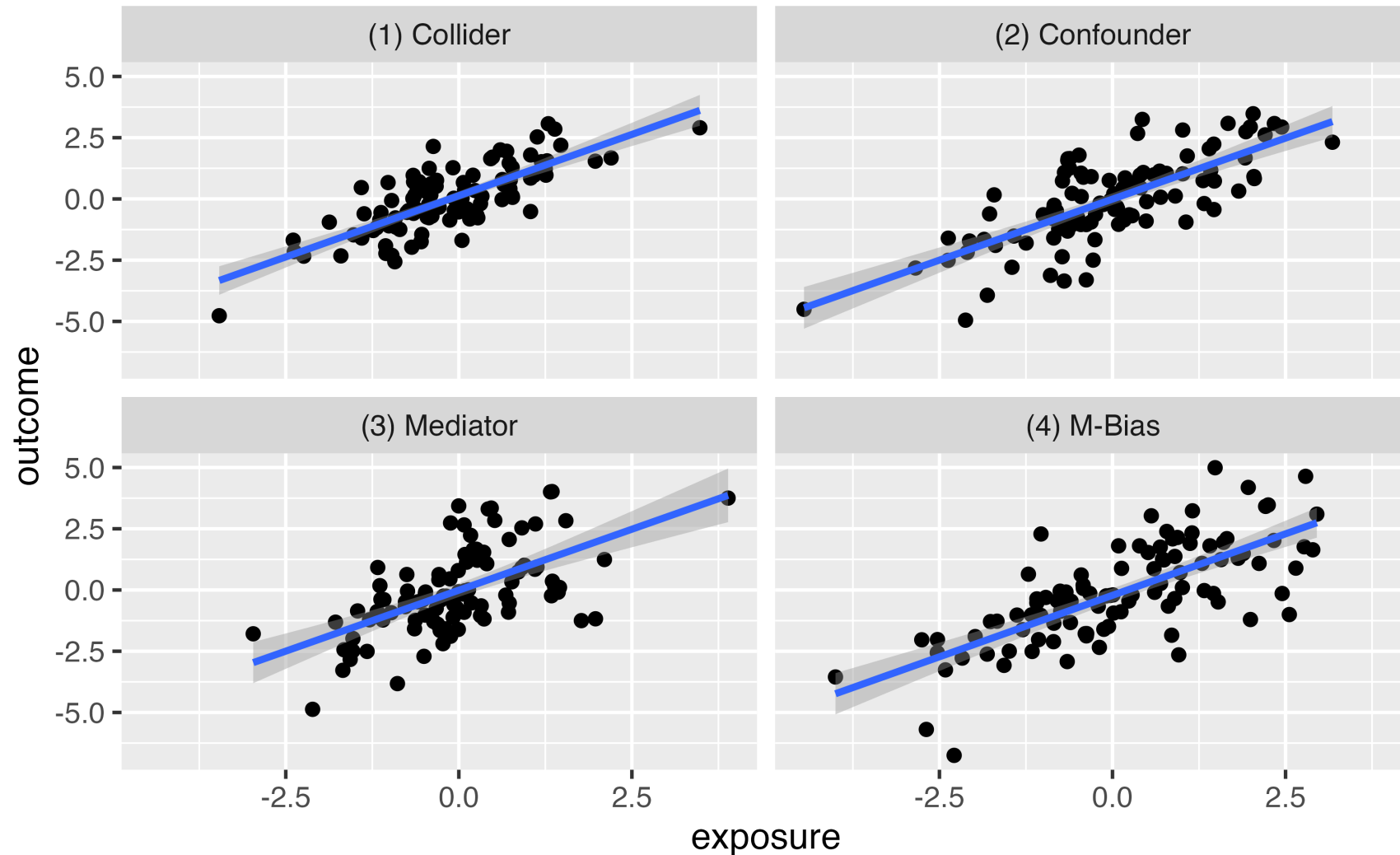
Load the **quartets** package

For each of the following 4 datasets, create a scatterplot looking at the relationship between **exposure** and **outcome**: **causal_collider**, **causal_confounding**, **causal_mediator**, **causal_m_bias**

For each of the above 4 datasets, look at the correlation between **exposure** and **covariate**

Stretch goal: For each of the above 4 datasets, fit a linear model to examine the relationship between the **exposure** and the **outcome**

Relationship between exposure and outcome



Relationship between exposure and covariate

```
1 causal_quartet |>  
2   group_by(dataset) |>  
3   summarise(corr = cor(exposure, covariate))
```

```
# A tibble: 4 × 2
```

	dataset	corr
	<chr>	<dbl>
1	(1) Collider	0.700
2	(2) Confounder	0.696
3	(3) Mediator	0.696
4	(4) M-Bias	0.696

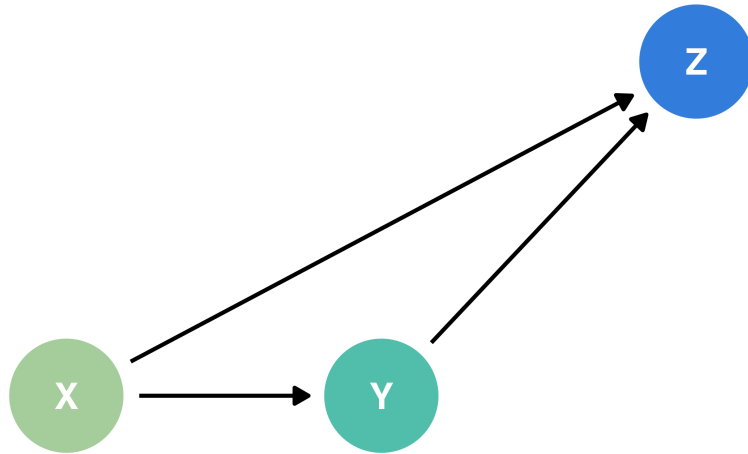
Observed effects

Data generating mechanism	ATE		
	ATE not adjusting for Z	adjusting for Z	Correlation of X and Z
(1) Collider	1.00	0.55	0.70
(2) Confounder	1.00	0.50	0.70
(3) Mediator	1.00	0.00	0.70
(4) M-Bias	1.00	0.88	0.70

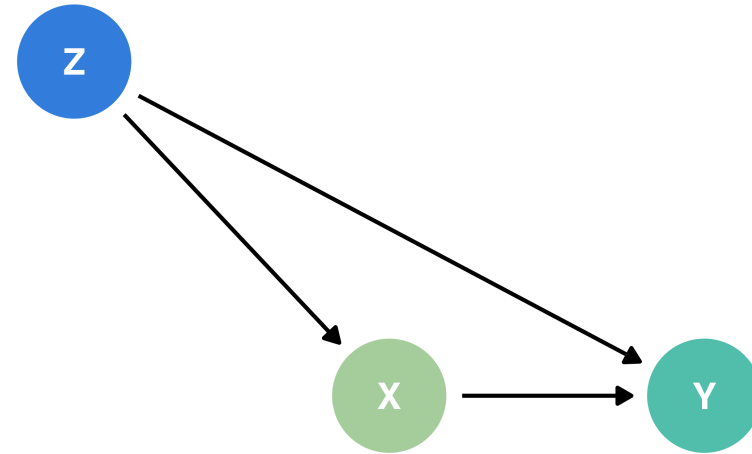
D’Agostino McGowan L, Gerke T, Barrett M (2023). Causal inference is not a statistical problem. Preprint arXiv:2304.02683v1.

The solution

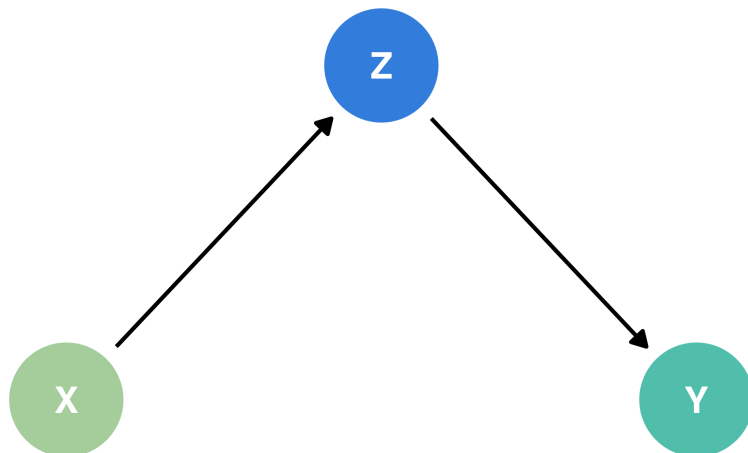
(1) Collider



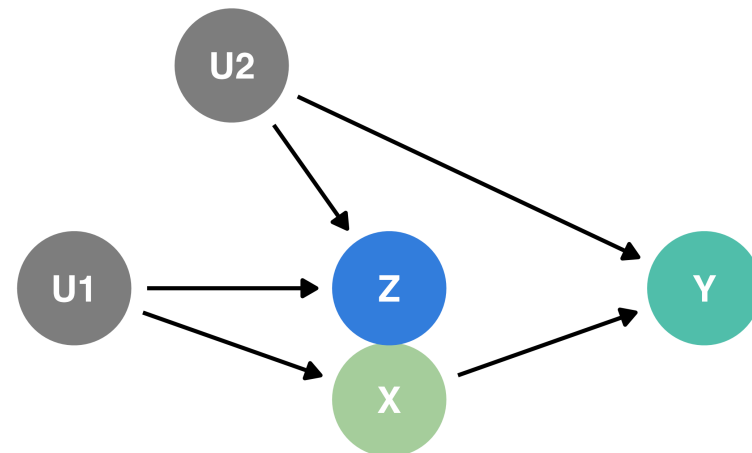
(2) Confounder



(3) Mediator



(4) M-bias



Correct effects

Data generating mechanism	Correct causal model	Correct causal effect
(1) Collider	$Y \sim X$	1.0
(2) Confounder	$Y \sim X ; Z$	0.5
(3) Mediator	Direct effect: $Y \sim X ; Z$ Total Effect: $Y \sim X$	Direct effect: 0.0 Total effect: 1.0
(4) M-Bias	$Y \sim X$	1.0

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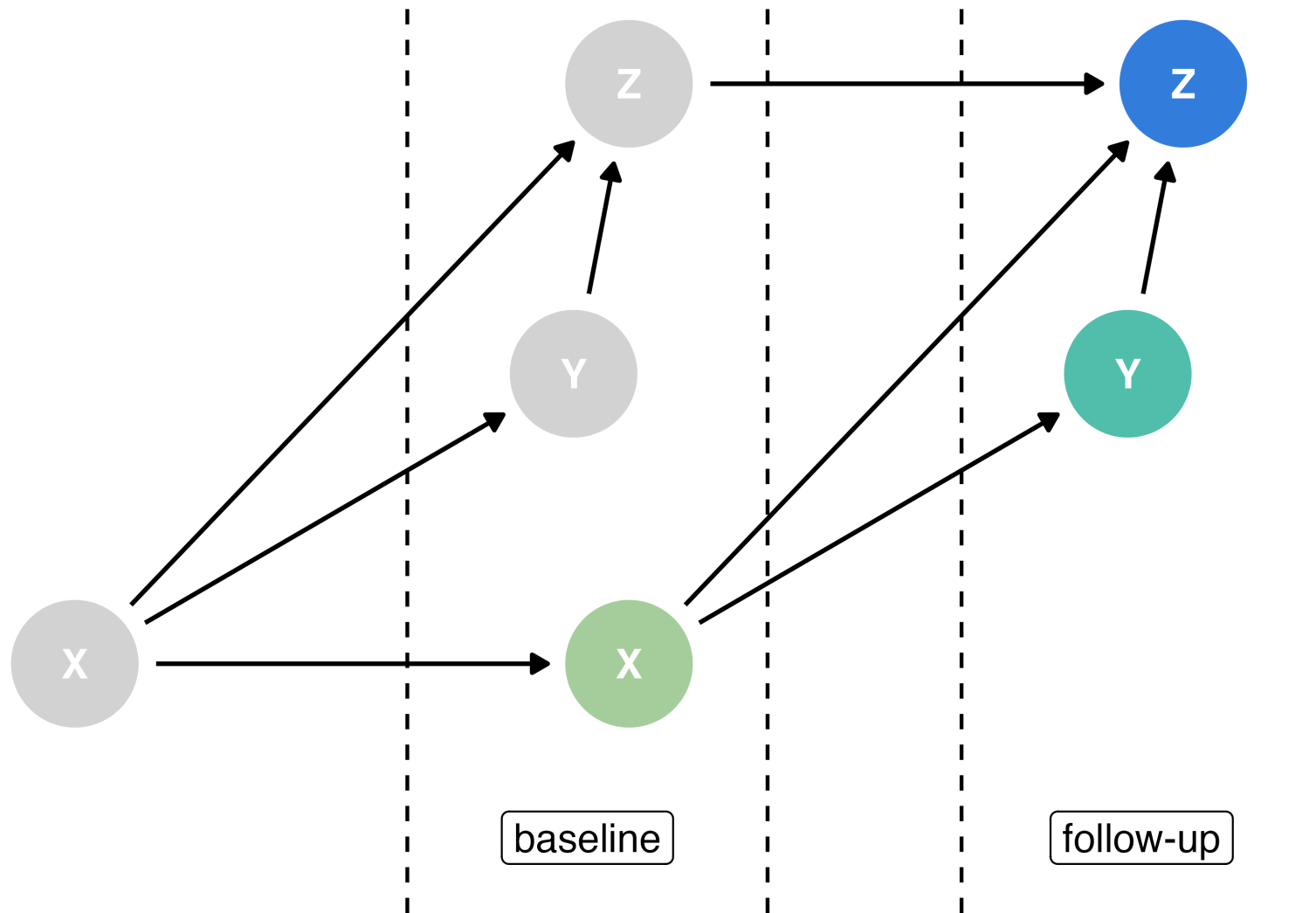
The *partial* solution

```
1 causal_collider_time
```

```
# A tibble: 100 × 6
  exposure_baseline outcome_baseline covariate_baseline
      <dbl>           <dbl>           <dbl>
1      -1.43         0.287         -0.0963
2       0.0593       -0.978         -1.11
3       0.370        0.348          0.647
4      0.00471      0.851          0.755
5       0.340        1.94          1.19
6      -3.61       -0.235        -0.588
7       1.44       -0.827        -1.13
8       1.02      -0.0410         0.689
9      -2.43       -2.10        -1.49
10     -1.26      -2.41        -2.78
# i 90 more rows
# i 3 more variables: exposure_followup <dbl>,
#   outcome_followup <dbl>, covariate_followup <dbl>
```

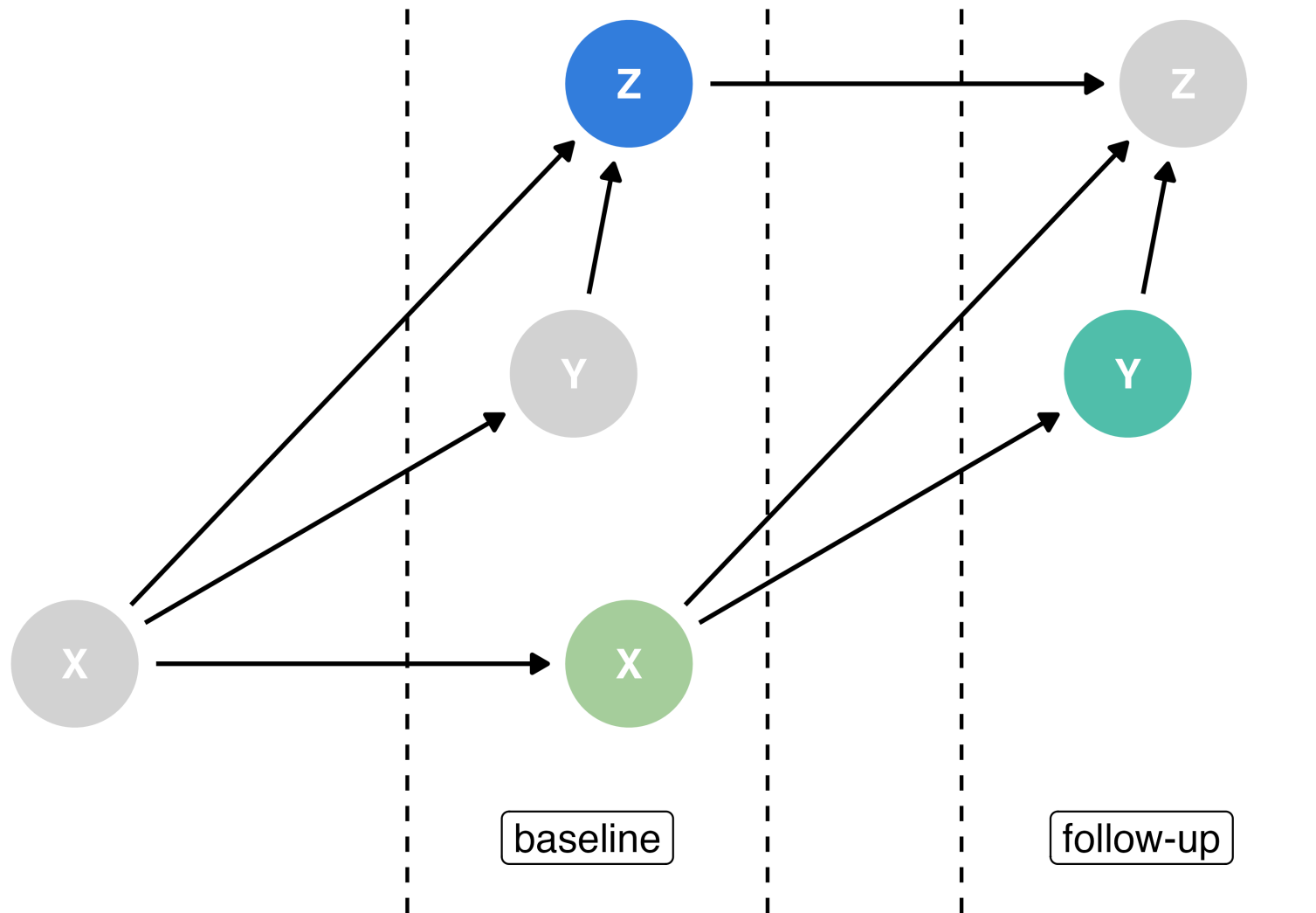
Time-varying data

Time-varying DAG



True causal effect: 1 Estimated causal effect: 0.55

Time-varying DAG



True causal effect: 1 Estimated causal effect: 1

```
outcome_followup ~ exposure_baseline +  
  covariate_baseline
```

Your turn 2

For each of the following 4 datasets, fit a linear model examining the relationship between **outcome_followup** and **exposure_baseline** adjusting for **covariate_baseline**:

- causal_collider_time,**
- causal_confounding_time,**
- causal_mediator_time, causal_m_bias_time**

The *partial* solution

Data generating mechanism	ATE not adjusting for pre-exposure Z	ATE adjusting for pre-exposure Z	Correct causal effect
(1) Collider	1.00	1.00	1.00
(2) Confounder	1.00	0.50	0.50
(3) Mediator	1.00	1.00	1.00
(4) M-Bias	1.00	0.88	1.00

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On *M*-Bias

- The relationship between Z and the unmeasured confounders needs to be really large (Liu et al 2012)
- “To obsess about the possibility of [M-bias] generates bad practical advice in all but the most unusual circumstances” (Rubin 2009)
- There are (almost) no true zeros (Gelman 2011)
- Asymptotic theory shows that induction of M-bias is quite sensitive to various deviations from the exact M-Structure (Ding and Miratrix 2014)